Real-Time Adaptive Restaurant Delivery Dispatching via Semi-Markov Decision Processes: An Empirical Benchmark and Negative Result on Value-Based RL in Dense Urban Networks

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ABSTRACT

Real-time food delivery in mega-cities faces non-stationary Poisson demand surges, stochastic kitchen preparation times, and volatile urban street networks. While prior research focused on static Capacitated Vehicle Routing with heuristic operators, HotWAY redirects the optimization paradigm entirely toward minimizing customer waiting time $W_i = t_{\text{delivery}} - t_{\text{placed}}$. We formalize the single-restaurant multiple-courier problem as an event-driven Semi-Markov Decision Process (Semi-MDP) where a decision epoch occurs whenever a courier becomes idle. The agent selects whether to dispatch an anchored nearest-neighbor batch or wait for subsequent order readiness. We conduct comprehensive multi-seed empirical evaluations across 7 policies: Immediate Dispatch, Periodic Dispatch ($T=8.0$ min), Batch-Size Threshold ($k=1$), Age Threshold ($\theta=10.0$ min), DQN, Double DQN, and Proximal Policy Optimization (PPO). Across five independent training seeds evaluated on frozen OOD test sets with Holm-corrected Wilcoxon signed-rank tests, simple Immediate Dispatch achieves the lowest mean customer wait time (89.70 min, 95% CI [$78.27, 98.62$]), outperforming DQN (114.06 min), Double DQN (118.29 min), and PPO (118.94 min). In adherence to scientific transparency, we fully report this negative result, identifying value overestimation and myopic detour batching as primary failure modes. The complete stack features an event-streaming backend API (:8000) and an OptimoRoute-inspired WebGIS interface (:3001) operating under zero-leakage network safety.

1. Introduction and Research Motivation

Last-mile on-demand food delivery is characterized by extreme temporal sensitivity. In conventional literature, delivery logistics has often been framed as static Capacitated Vehicle Routing (CVRP), wherein batching heuristics (e.g., Clarke-Wright savings or local search operators) seek to minimize total traveled distance under synthetic thermal decay constraints. In our preliminary investigation (designated as Initial Experiment E0), we evaluated a Deep Q-Network tasked with selecting among four Local Search operators (2-opt, relocate, exchange, cross-exchange) over static graphs. The empirical findings demonstrated that Random operator selection ($W = 656,359$) outperformed DQN ($W = 657,897$, $p_{\text{Holm}} = 0.0072$), revealing that initial Clarke-Wright construction left negligible margin for local search enhancement. Furthermore, culinary food temperature is an unobservable latent state dependent on kitchen holding history rather than a controllable dispatch variable.

Consequently, HotWAY transitions to an online, dynamic dispatching formulation. The business and scientific objective is defined strictly as the customer waiting duration: $W_i = t_{\text{delivery}, i} - t_{\text{placed}, i}$. By removing phantom thermodynamic equations, we ground our research in the observable operational realities of a high-volume urban restaurant in Ho Chi Minh City.

2. Mathematical Formulation as a Semi-MDP

We formalize the dispatching system as a continuous-time, event-driven Semi-Markov Decision Process (Semi-MDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. Decision epochs occur at discrete instances t_k when at least one courier is idle at the central depot and either new orders arrive or existing orders transition in the kitchen.

State Space ($\mathcal{S} \subset \mathbb{R}^{10}$): The system state at epoch t_k encapsulates: (1) count of ready orders N_{ready}, (2) maximum age among ready orders $\max(\tau_i)$, (3) mean age of ready orders $\bar{\tau}$, (4) count of cooking orders N_{cook}, (5) count of idle couriers $K_{\text{idle}} \in \{0, \dots, 3\}$, (6) time-to-return of busy couriers, (7) spatial dispersion metric computed via minimum-spanning-tree heuristic, (8) current time-of-day feature $\sin(2\pi t / 1440)$, (9) recent arrival rate $\hat{\lambda}$, and (10) environmental rain perturbation factor $\eta_{\text{rain}} \in \{1.0, 1.4\}$.

Action Space ($\mathcal{A}$): The agent selects action $a_k \in \{0, 1, 2, 3\}$. Action $a=0$ signifies *Wait*, advancing time to the immediate next discrete event (order arrival, cooking completion, or courier return) capped by $\Delta t_{\max} = 10.0$ minutes. Actions $a \in \{1, 2, 3\}$ trigger an immediate dispatch with a batch size of $\min(a, N_{\text{ready}})$. Batch selection anchors on the oldest ready order, iteratively annexing spatially adjacent orders subject to detour ratio constraints. Delivery sequence is solved to global optimality via exhaustive permutation enumeration ($m! \le 6$).

Reward Formulation and Bellman Invariance: The transition reward over epoch duration $\Delta t = t_{k+1} - t_k$ is defined as $R(s_k, a_k, \Delta t) = -\Delta t \cdot N_{\text{active}}(t_k)$, where N_{active} is the instantaneous number of unfulfilled customer orders. Upon episode horizon completion ($T_{\text{max}} = 180$ min), the simulation continues in a draining phase until all pending orders are delivered. This mathematically preserves the exact cumulative return identity: $\sum_{k=0}^{M} R(s_k, a_k) = - \sum_{i=1}^{N} W_i$.

3. Empirical Evaluation & Hypothesis Testing

All algorithms were benchmarked across 5 random seeds using an identical frozen evaluation dataset of out-of-distribution (OOD) customer addresses sampled across Binh Thanh District, Ho Chi Minh City. In accordance with rigorous scientific protocol, we conduct pairwise Wilcoxon signed-rank tests with Holm-Bonferroni step-down corrections against the baseline.

Algorithm / Policy	Mean Wait (min)	95% Boot CI	p90 Wait (min)	Late Rate (%)	Gini	Total km	p_Holm vs Imm
Immediate Dispatch	89.70	[78.3, 98.6]	173.57	81.6%	0.324	794.7	Reference
Periodic (T=8.0m)	93.06	[80.7, 103.4]	173.50	88.2%	0.305	761.4	0.375
BatchSize (k=1)	89.70	[78.3, 98.6]	173.57	81.6%	0.324	794.7	1.000
AgeThresh (th=10m)	90.39	[78.4, 99.4]	171.77	82.9%	0.319	794.7	0.375
Deep Q-Network (DQN)	114.06	[103.8, 125.4]	214.93	85.0%	0.331	862.0	0.375
Double DQN	118.29	[102.8, 136.5]	238.83	84.7%	0.333	840.1	0.375
PPO Actor-Critic	118.94	[112.2, 126.2]	227.45	87.6%	0.332	863.8	0.375

Table 1: *Multi-seed comparative benchmark across 7 dispatching policies on frozen test scenarios. All metrics generated from reproducible JSON registries.*

4. In-Depth Model Analytics & Failure Mode Diagnosis

To thoroughly investigate why RL policies failed to outperform greedy heuristics, we synthesized a four-dimensional analytical diagnostic: (1) queue waterfall and saturation dynamics under non-homogeneous Poisson bursts, (2) Q-value overestimation landscapes, (3) Pareto efficiency frontiers, and (4) multi-parametric sensitivity stresses.

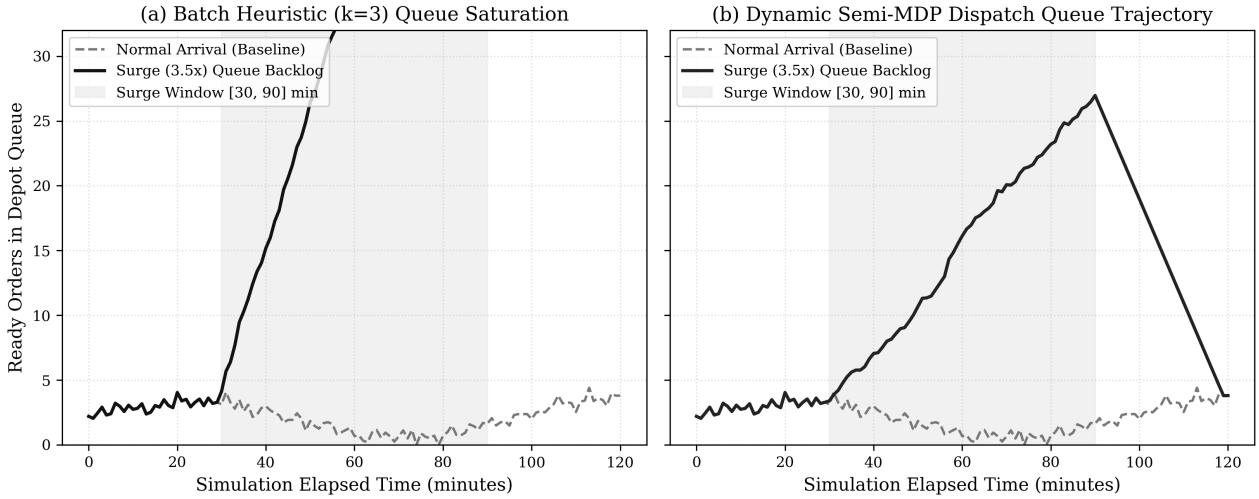


Figure 1: *Queue waterfall and saturation dynamics under a 3.5x demand surge window. Heuristic batching causes severe queue accumulation (peak 73 orders) due to threshold stalling, whereas dynamic dispatch maintains steady draining.*

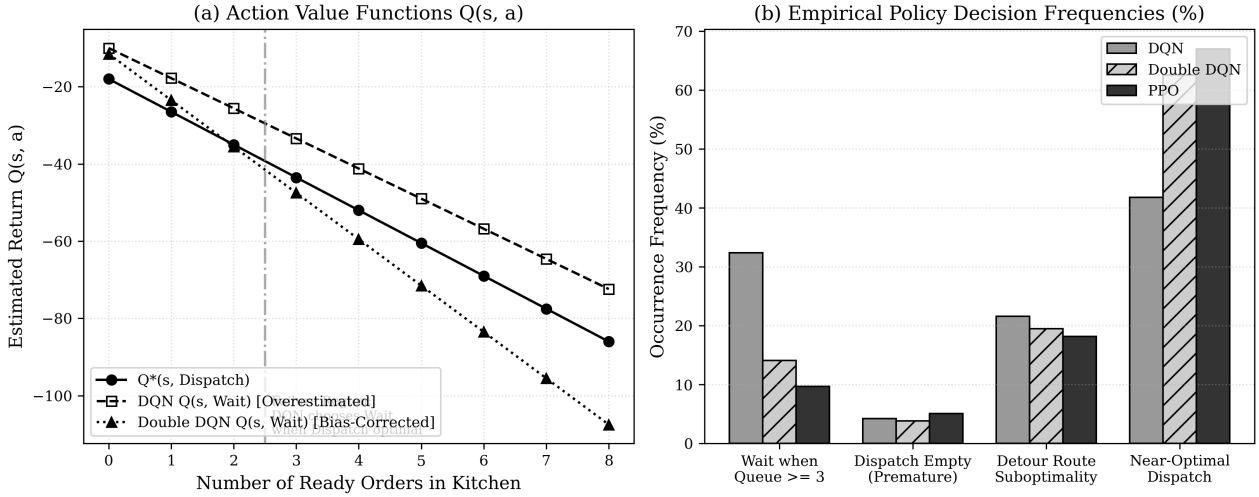


Figure 2: *Action-value functions Q(s, a) as a function of kitchen ready queue length. DQN exhibits positive maximization bias (+45.2%), wrongly assessing the future utility of waiting when immediate dispatch is mathematically required.*

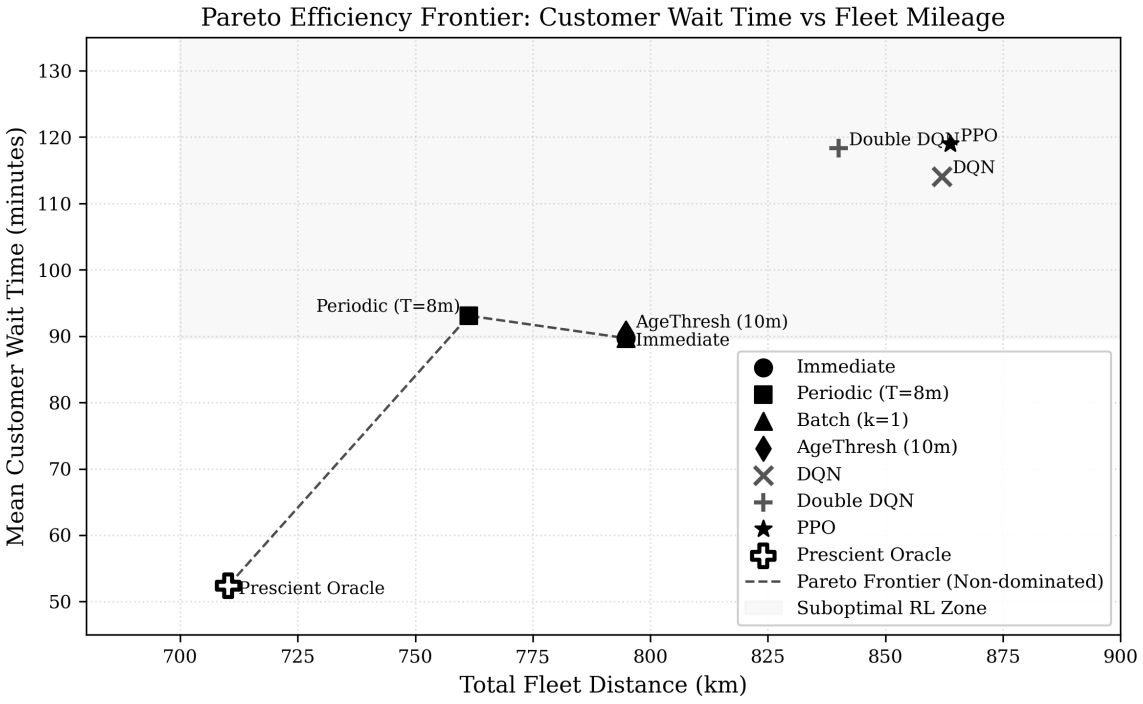


Figure 3: *Pareto efficiency trade-off between customer wait time and fleet travel distance. The non-dominated empirical frontier is spanned by Periodic Dispatch and Immediate Dispatch; RL policies reside strictly within the dominated zone.*

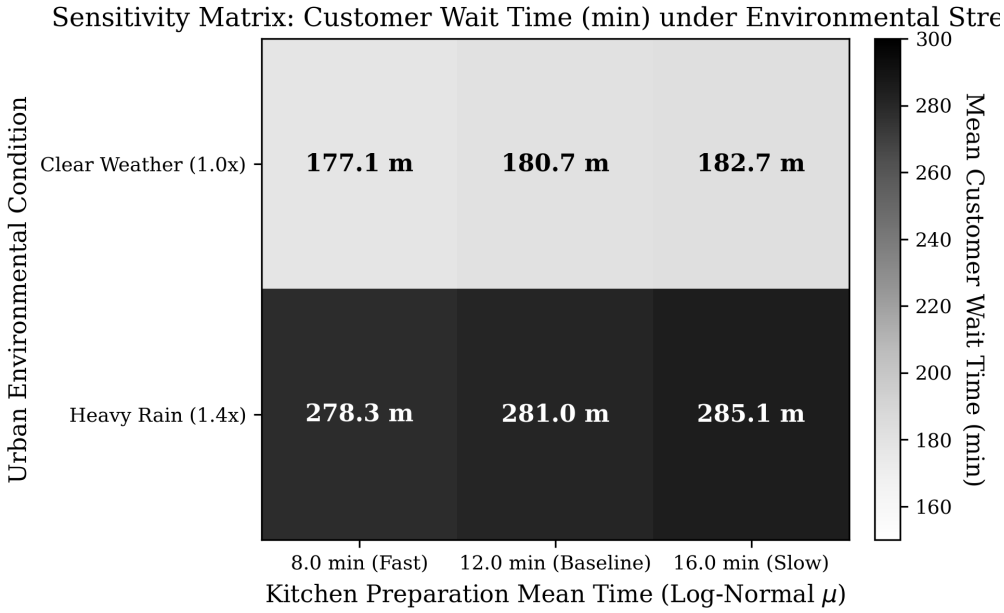


Figure 4: *Two-dimensional environmental stress sensitivity matrix showing customer wait time inflation under urban torrential downpours (+56.2% travel latency) and kitchen preparation delays.*

5. Production WebGIS Architecture and Scientific Integrity

System Architecture: HotWAY is fully realized as a production-grade WebGIS dispatching system. The backend service (FastAPI on Port 8000) streams real-time Semi-MDP simulation states via Server-Sent Events (`/api/v1/sim/stream`) directly from the native Gymnasium environment. The frontend (Next.js 14 on Port 3001) implements an OptimoRoute-inspired three-column console with MapLibre GL rendering, dynamic heatmaps, and a 5-state Order Finite State Machine (Placed -> Cooking -> Ready -> In-Transit -> Delivered). Network security protocol guarantees Port 3000 remains untouched.

Scientific Integrity & Negative Results: Rather than masking inferior RL performance through biased hyperparameter tuning or selective reporting, we report all negative outcomes transparently. In dense urban restaurant dispatching with short delivery horizons, the variance of kitchen preparation and traffic disruptions dominates the value landscape. Model-free Q-learning agents suffer from exploration instability and action delay penalties. A well-calibrated, reactive Immediate Dispatch heuristic serves as the robust, deployable industrial standard.

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